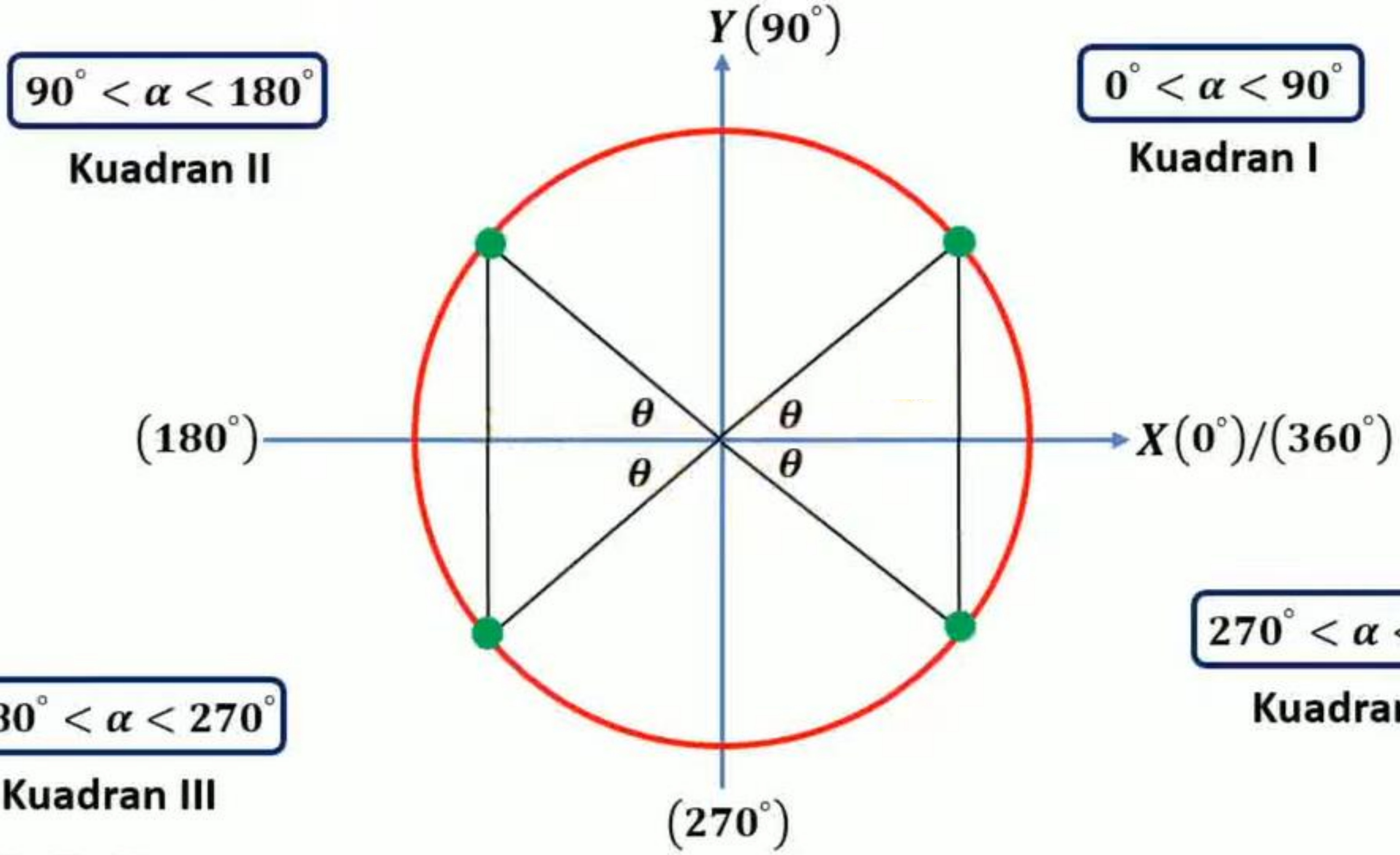
SUDUT BERELASI TRIGONOMETRI

TRIGONOMETRI

Sudut – Sudut Berelasi

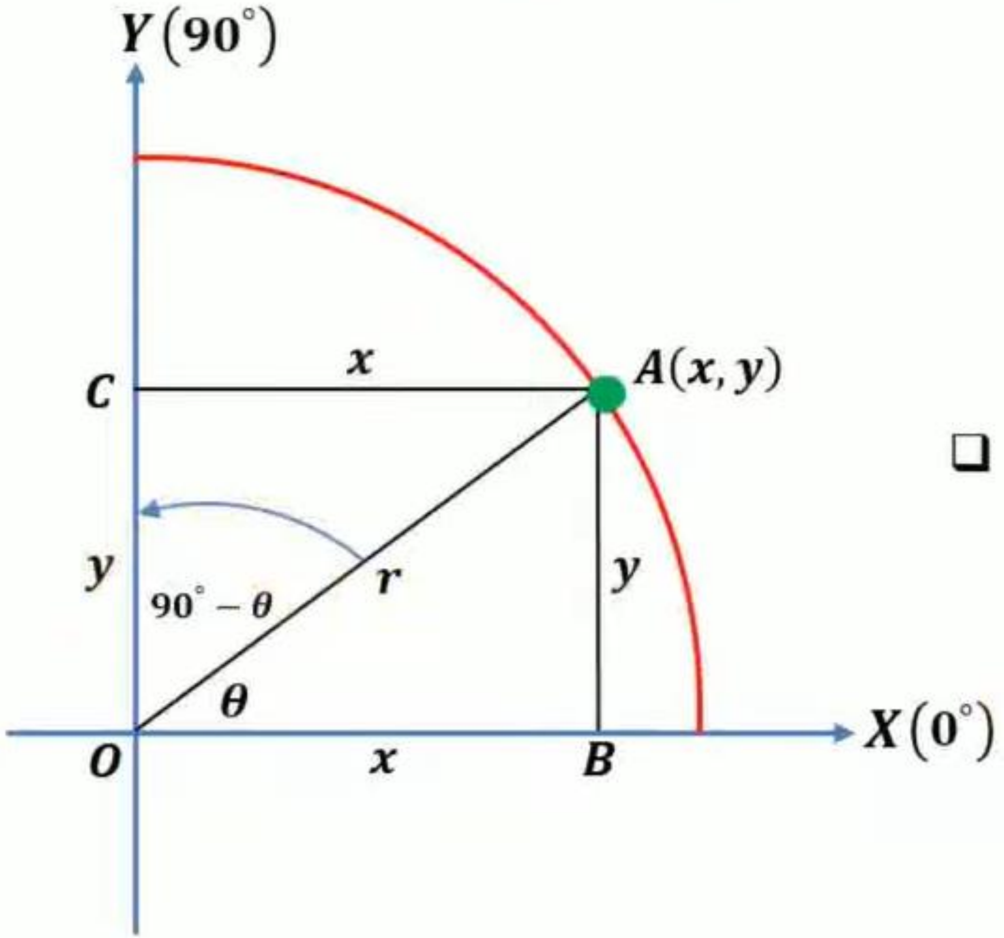
1. Pembagian Sudut dalam Trigonometri



TRIGONOMETRI

Sudut – Sudut Berelasi

2. Perbandingan Trigonometri Sudut di Kuadran I



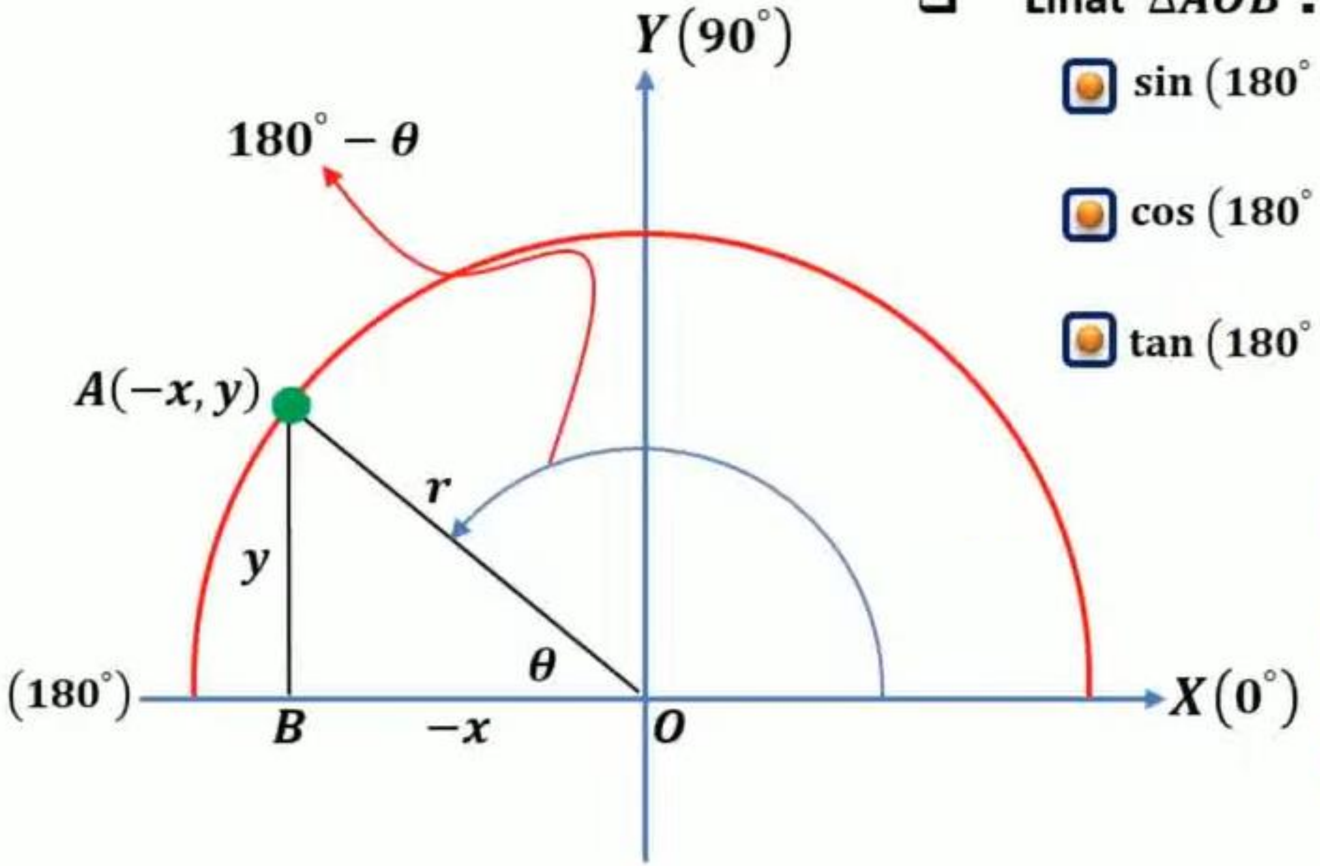
- Lihat ΔAOB :
  - $\sin\theta = \dfrac{De}{Mi} = \dfrac{y}{r}$ ⟷ $\operatorname{cosec}\theta = \dfrac{Mi}{De} = \dfrac{r}{y}$
  - $\cos\theta = \dfrac{Sa}{Mi} = \dfrac{x}{r}$ ⟷ $\sec\theta = \dfrac{Mi}{Sa} = \dfrac{r}{x}$
  - $\tan\theta = \dfrac{De}{Sa} = \dfrac{y}{x}$ ⟷ $\cot\theta = \dfrac{Sa}{De} = \dfrac{x}{y}$

- Lihat ΔAOC :
  - $\sin(90^\circ - \theta) = \dfrac{De}{Mi} = \dfrac{x}{r} = \cos\theta$
  - $\operatorname{cosec}(90^\circ - \theta) = \sec\theta$
  - $\cos(90^\circ - \theta) = \dfrac{Sa}{Mi} = \dfrac{y}{r} = \sin\theta$
  - $\sec(90^\circ - \theta) = \operatorname{cosec}\theta$
  - $\tan(90^\circ - \theta) = \dfrac{De}{Sa} = \dfrac{x}{y} = \cot\theta$
  - $\cot(90^\circ - \theta) = \tan\theta$

TRIGONOMETRI

Sudut – Sudut Berelasi

3. Perbandingan Trigonometri Sudut di Kuadran II



- Lihat ΔAOB :

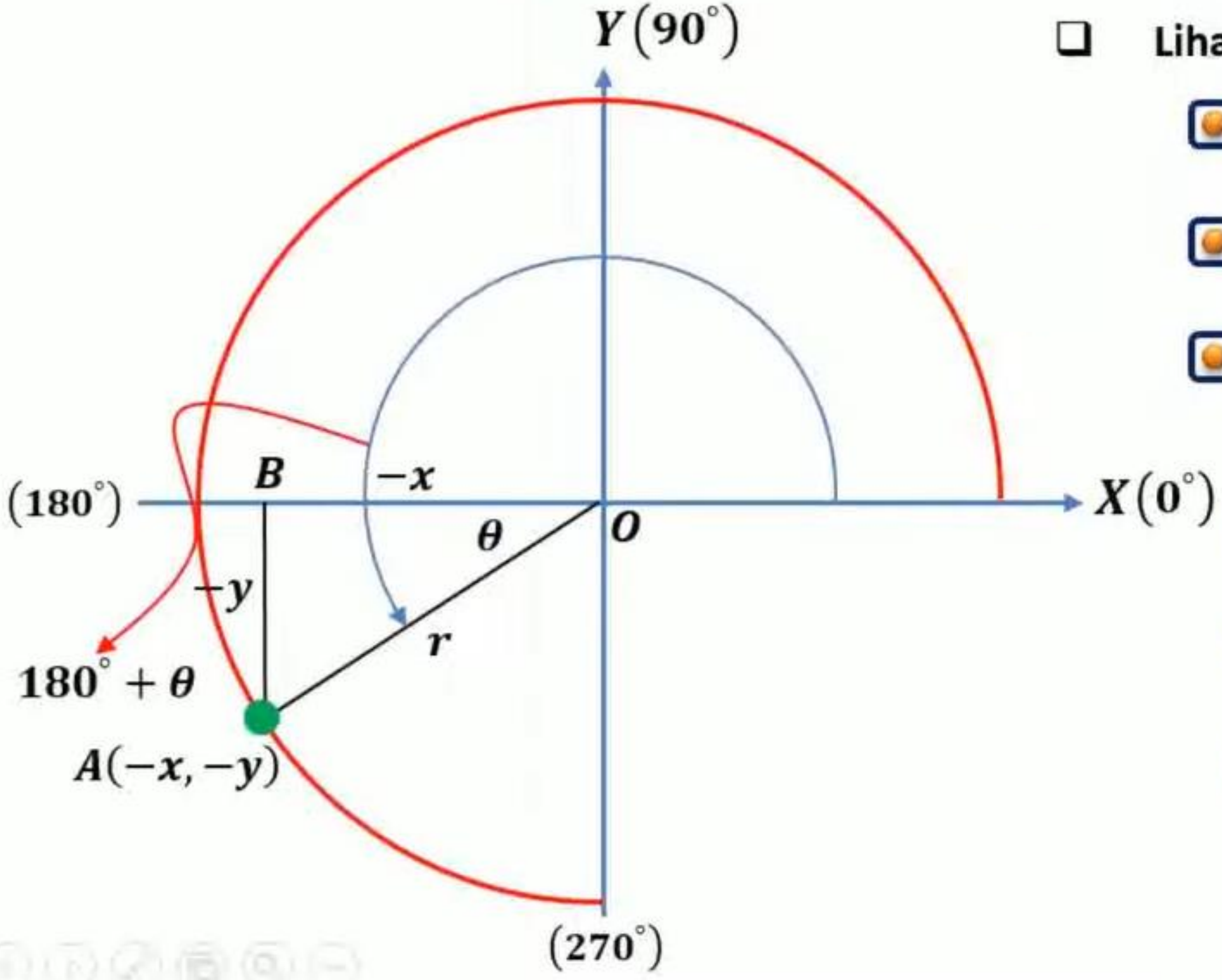
  - $\sin(180^\circ - \theta) = \frac{De}{Mi} = \frac{y}{r} = \sin\theta$
  - $\cos(180^\circ - \theta) = \frac{Sa}{Mi} = \frac{-x}{r} = -\frac{x}{r} = -\cos\theta$
  - $\tan(180^\circ - \theta) = \frac{De}{Sa} = \frac{y}{-x} = -\frac{y}{x} = -\tan\theta$

  - $\text{cosec}\,(180^\circ - \theta) = \frac{Mi}{De} = \frac{r}{y} = \text{cosec}\,\theta$
  - $\sec(180^\circ - \theta) = \frac{Mi}{Sa} = \frac{r}{-x} = -\frac{r}{x} = -\sec\theta$
  - $\cot(180^\circ - \theta) = \frac{Sa}{De} = \frac{-x}{y} = -\frac{x}{y} = -\cot\theta$

TRIGONOMETRI

Sudut – Sudut Berelasi

4. Perbandingan Trigonometri Sudut di Kuadran III

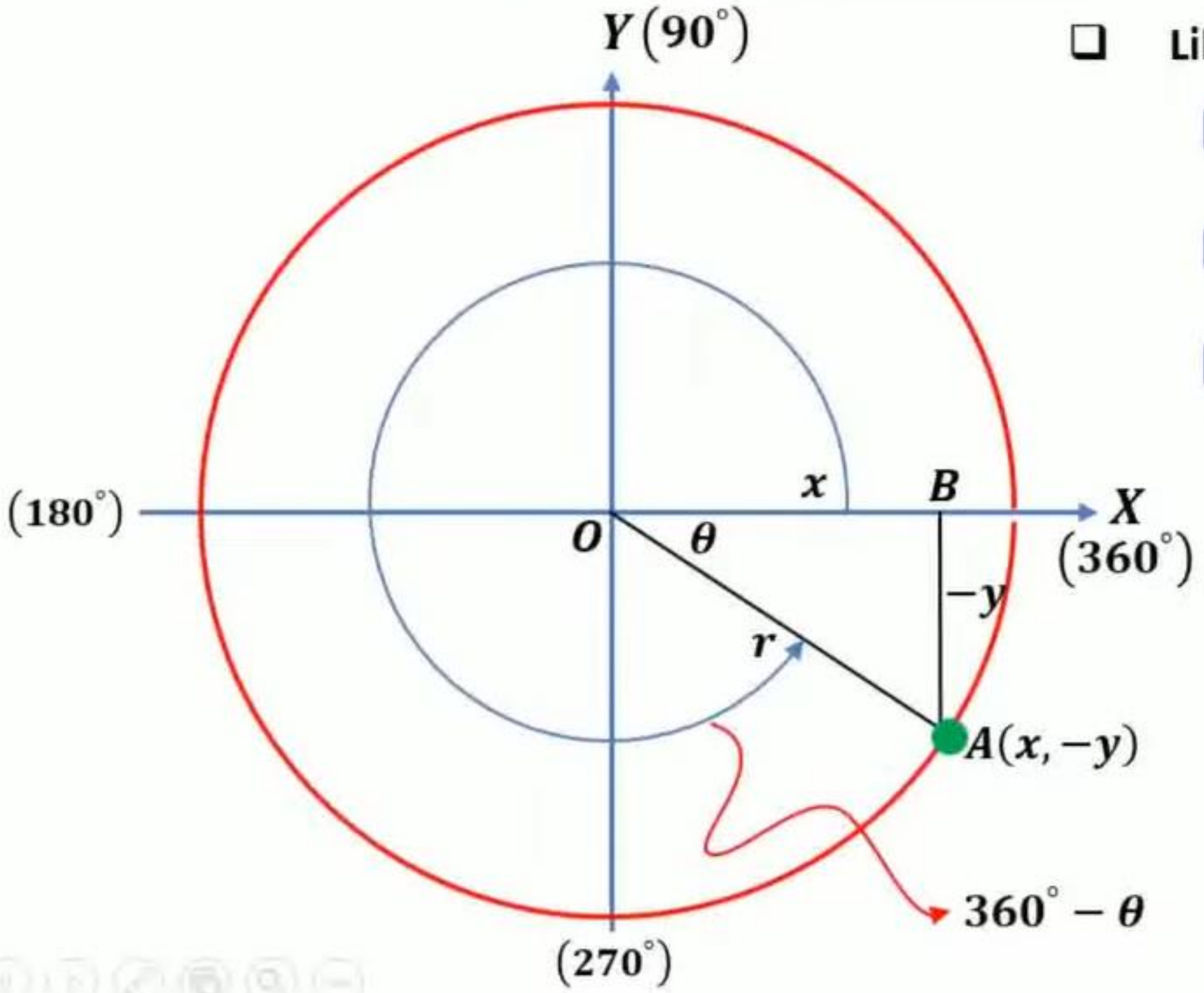


- Lihat ΔAOB :
  - $\sin(180^\circ + \theta) = \dfrac{De}{Mi} = \dfrac{-y}{r} = -\sin\theta$
  - $\cos(180^\circ + \theta) = \dfrac{Sa}{Mi} = \dfrac{-x}{r} = -\dfrac{x}{r} = -\cos\theta$
  - $\tan(180^\circ + \theta) = \dfrac{De}{Sa} = \dfrac{-y}{-x} = \dfrac{y}{x} = \tan\theta$
  - $\mathrm{cosec}(180^\circ + \theta) = \dfrac{Mi}{De} = \dfrac{r}{-y} = -\dfrac{r}{y} = -\mathrm{cosec}\,\theta$
  - $\sec(180^\circ + \theta) = \dfrac{Mi}{Sa} = \dfrac{r}{-x} = -\dfrac{r}{x} = -\sec\theta$
  - $\cot(180^\circ + \theta) = \dfrac{Sa}{De} = \dfrac{-x}{-y} = \dfrac{x}{y} = \cot\theta$

TRIGONOMETRI

Sudut – Sudut Berelasi

5. Perbandingan Trigonometri Sudut di Kuadran IV



- Lihat ΔAOB :

$$\sin(360^\circ - \theta) = \frac{De}{Mi} = \frac{-y}{r} = -\sin\theta$$

$$\cos(360^\circ - \theta) = \frac{Sa}{Mi} = \frac{x}{r} = \cos\theta$$

$$\tan(360^\circ - \theta) = \frac{De}{Sa} = \frac{-y}{x} = -\frac{y}{x} = -\tan\theta$$

$$\operatorname{cosec}(360^\circ - \theta) = \frac{Mi}{De} = \frac{r}{-y} = -\frac{r}{y} = -\operatorname{cosec}\theta$$

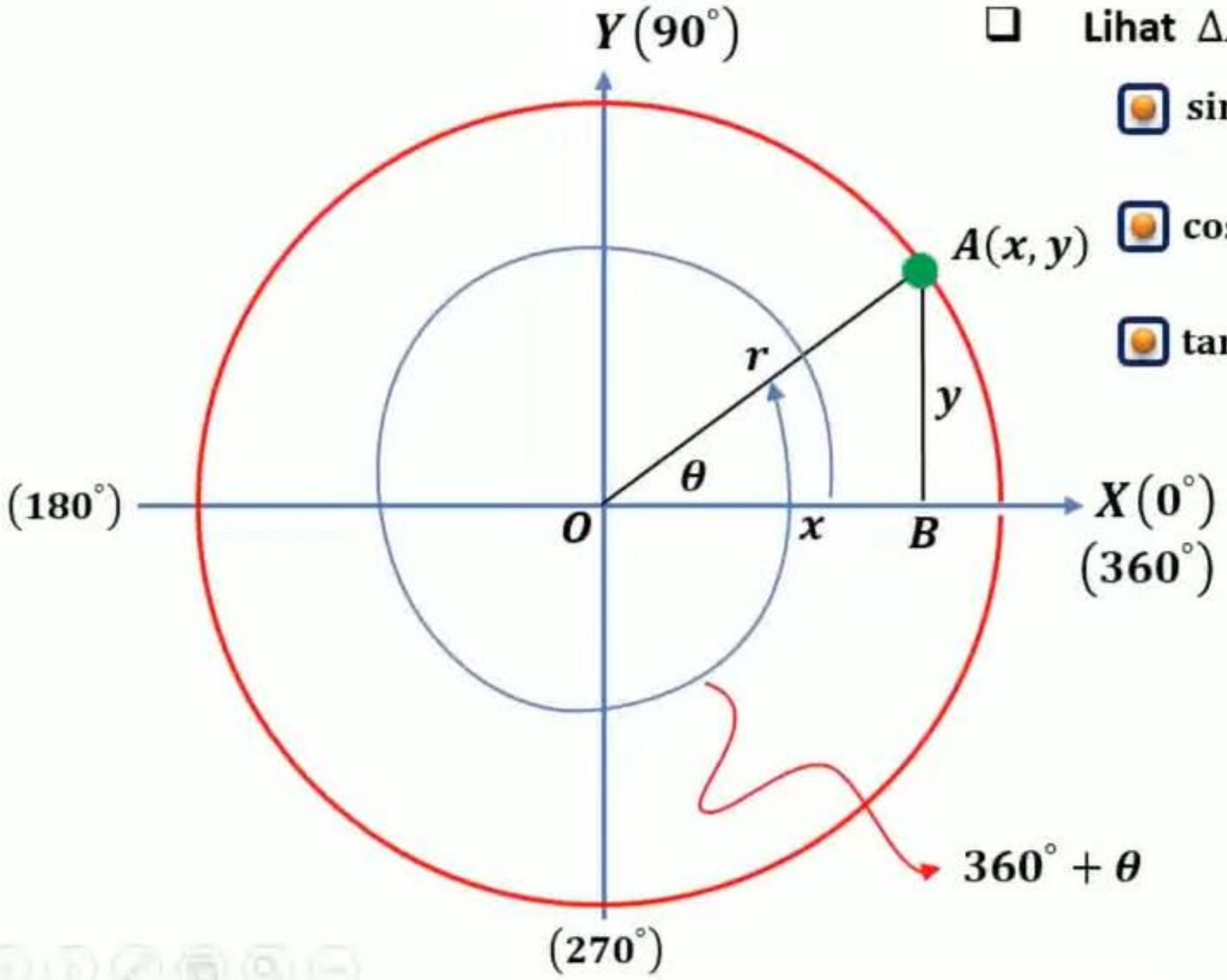
$$\sec(360^\circ - \theta) = \frac{Mi}{Sa} = \frac{r}{x} = \sec\theta$$

$$\cot(360^\circ - \theta) = \frac{Sa}{De} = \frac{x}{-y} = -\frac{x}{y} = -\cot\theta$$

TRIGONOMETRI

Sudut – Sudut Berelasi

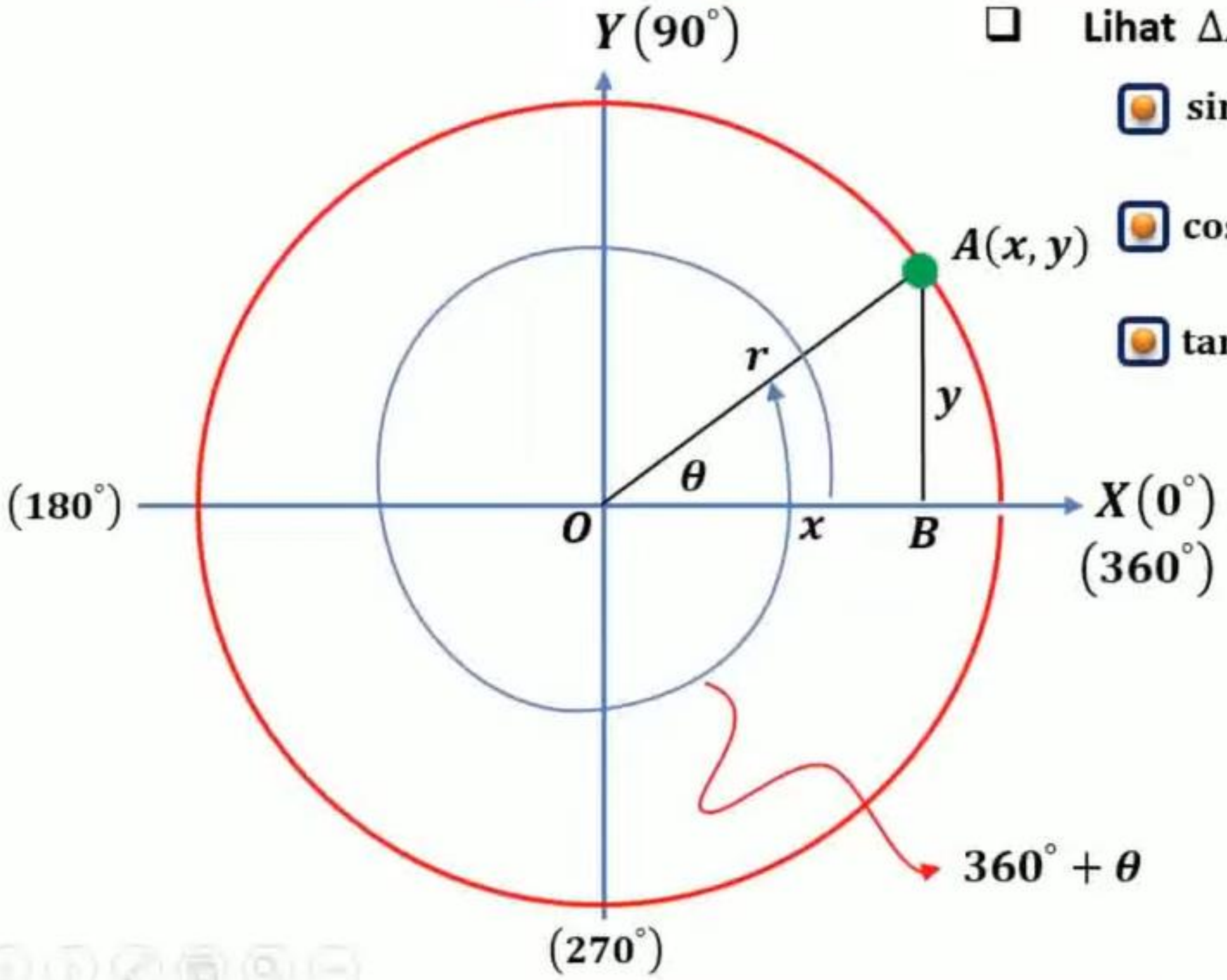
6. Perbandingan Trigonometri Sudut Lebih dari 360° atau Negatif



- Lihat ΔAOB :
  - $\sin(k \cdot 360^\circ + \theta) = \sin\theta$
  - $\text{cosec}(k \cdot 360^\circ + \theta) = \text{cosec}\,\theta$
  - $\cos(k \cdot 360^\circ + \theta) = \cos\theta$
  - $\sec(k \cdot 360^\circ + \theta) = \sec\theta$
  - $\tan(k \cdot 360^\circ + \theta) = \tan\theta$
  - $\cot(k \cdot 360^\circ + \theta) = \cot\theta$

Sudut – Sudut Berelasi

6. Perbandingan Trigonometri Sudut Lebih dari 360° atau Negatif



- Lihat ΔAOB :

$\sin(k \cdot 360^\circ + \theta) = \sin\theta$ | $\text{cosec}(k \cdot 360^\circ + \theta) = \text{cosec}\,\theta$

$\cos(k \cdot 360^\circ + \theta) = \cos\theta$ | $\sec(k \cdot 360^\circ + \theta) = \sec\theta$

$\tan(k \cdot 360^\circ + \theta) = \tan\theta$ | $\cot(k \cdot 360^\circ + \theta) = \cot\theta$

Relasi Sudut di Kuadran I

$\sin(90^\circ - \theta) = \cos\theta$

$\cos(90^\circ - \theta) = \sin\theta$

$\tan(90^\circ - \theta) = \cot\theta$

$\cot(90^\circ - \theta) = \tan\theta$

$\sec(90^\circ - \theta) = \csc\theta$

$\csc(90^\circ - \theta) = \sec\theta$

$\sin(360^\circ + \theta) = \sin\theta$

$\cos(360^\circ + \theta) = \cos\theta$

$\tan(360^\circ + \theta) = \tan\theta$

$\cot(360^\circ + \theta) = \cot\theta$

$\sec(360^\circ + \theta) = \sec\theta$

$\csc(360^\circ + \theta) = \csc\theta$

Relasi Sudut di Kuadran II

$\sin(90^\circ + \theta) = \cos\theta$

$\cos(90^\circ + \theta) = -\sin\theta$

$\tan(90^\circ + \theta) = -\cot\theta$

$\cot(90^\circ + \theta) = -\tan\theta$

$\sec(90^\circ + \theta) = -\csc\theta$

$\csc(90^\circ + \theta) = \sec\theta$

$\sin(180^\circ - \theta) = \sin\theta$

$\cos(180^\circ - \theta) = -\cos\theta$

$\tan(180^\circ - \theta) = -\tan\theta$

$\cot(180^\circ - \theta) = -\cot\theta$

$\sec(180^\circ - \theta) = -\sec\theta$

$\csc(180^\circ - \theta) = \csc\theta$

Relasi Sudut di Kuadran III

$\sin(180^\circ + \theta) = -\sin\theta$

$\cos(180^\circ + \theta) = -\cos\theta$

$\tan(180^\circ + \theta) = \tan\theta$

$\cot(180^\circ + \theta) = \cot\theta$

$\sec(180^\circ + \theta) = -\sec\theta$

$\csc(180^\circ + \theta) = -\csc\theta$

$\sin(270^\circ - \theta) = -\cos\theta$

$\cos(270^\circ - \theta) = -\sin\theta$

$\tan(270^\circ - \theta) = \cot\theta$

$\cot(270^\circ - \theta) = \tan\theta$

$\sec(270^\circ - \theta) = -\csc\theta$

$\csc(270^\circ - \theta) = -\sec\theta$

Relasi Sudut di Kuadran IV

$\sin(270^\circ + \theta) = -\cos\theta$

$\cos(270^\circ + \theta) = \sin\theta$

$\tan(270^\circ + \theta) = -\cot\theta$

$\cot(270^\circ + \theta) = -\tan\theta$

$\sec(270^\circ + \theta) = \csc\theta$

$\csc(270^\circ + \theta) = -\sec\theta$

$\sin(360^\circ - \theta) = -\sin\theta$

$\cos(360^\circ - \theta) = \cos\theta$

$\tan(360^\circ - \theta) = -\tan\theta$

$\cot(360^\circ - \theta) = -\cot\theta$

$\sec(360^\circ - \theta) = \sec\theta$

$\csc(360^\circ - \theta) = -\csc\theta$

Relasi Sudut $(-\theta)$

$\sin(-\theta) = -\sin\theta$

$\cos(-\theta) = \cos\theta$

$\tan(-\theta) = -\tan\theta$

$\cot(-\theta) = -\cot\theta$

$\sec(-\theta) = \sec\theta$

$\csc(-\theta) = -\csc\theta$

Relasi Sudut Lebih dari 360°

$\sin(\theta + k.360^\circ) = \sin\theta$

$\cos(\theta + k.360^\circ) = \cos\theta$

$\tan(\theta + k.360^\circ) = \tan\theta$

$\cot(\theta + k.360^\circ) = \cot\theta$

$\sec(\theta + k.360^\circ) = \sec\theta$

$\csc(\theta + k.360^\circ) = \csc\theta$

Contoh 1

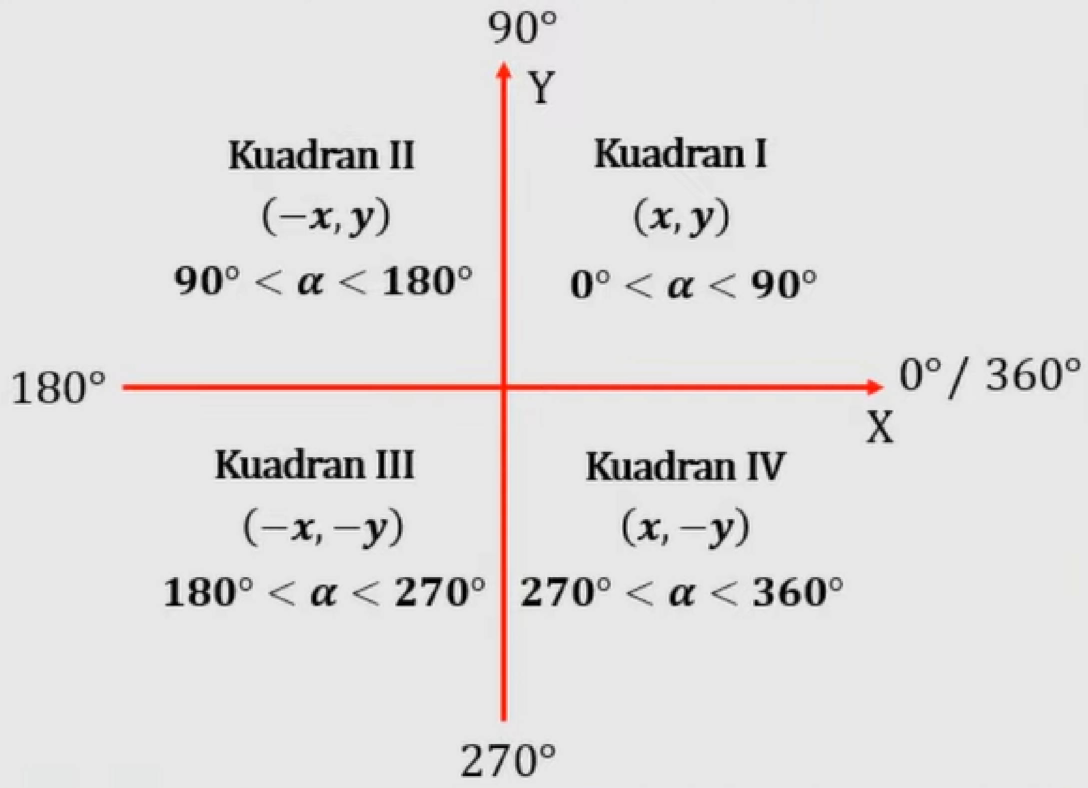
Tentukan nilai trigonometri sudut-sudut berikut ini:

a. $\sin 135^\circ$
b. $\cos 210^\circ$
c. $\tan 315^\circ$
d. $\sec 300^\circ$

Penyelesaian

a. $\sin 135^\circ = \sin(180^\circ - 45^\circ)$
$$= \sin 45^\circ$$
$$= \frac{1}{2}\sqrt{2}$$

Letak sudut pada bidang cartesius



Tanda nilai trigonometri di berbagai kuadran

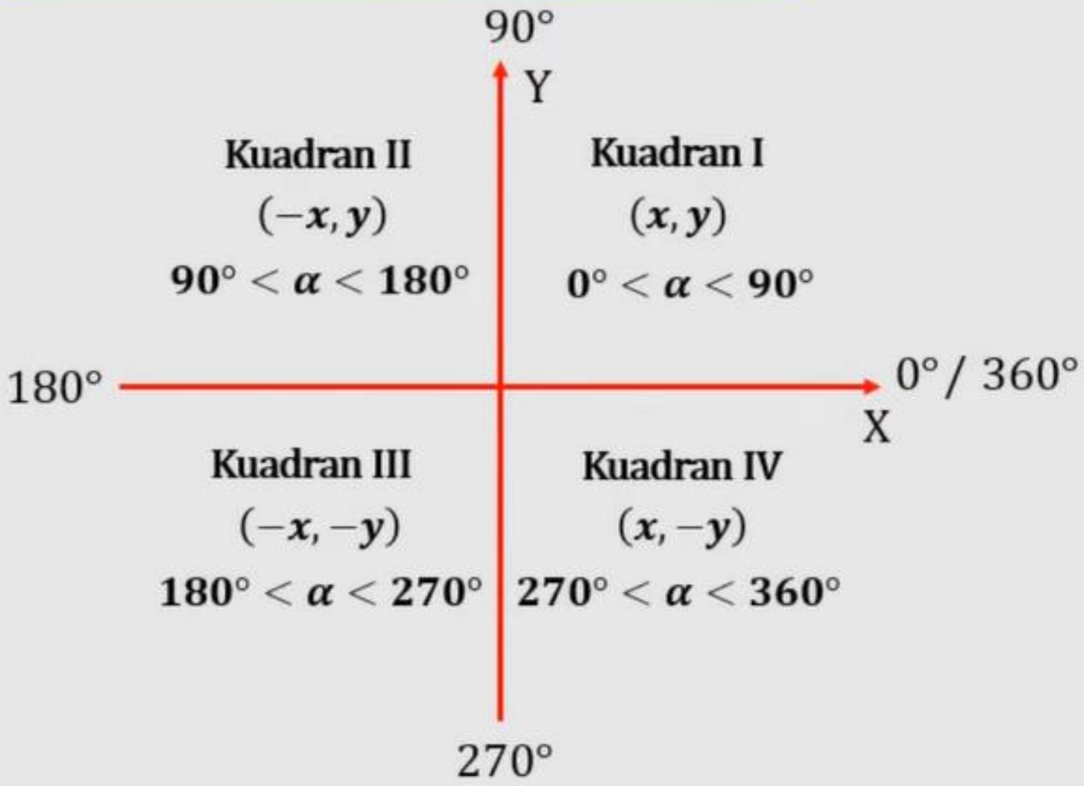
"Semua Sindikat Tangannya Kosong

I II III IV

$$\text{b. } \cos 210^\circ = -\cos(180^\circ + 30^\circ)$$
$$= -\cos 30^\circ$$
$$= -\frac{1}{2}\sqrt{3}$$

$$\text{c. } \tan 315^\circ = -\tan(360^\circ - 45^\circ)$$
$$= -\tan 45^\circ$$
$$= -1$$

Letak sudut pada bidang cartesius



Tanda nilai trigonometri di berbagai kuadran

"Semua Sindikat Tangannya Kosong

I II III IV

b. $\cos 210^\circ = -\cos(180^\circ + 30^\circ)$

$$= -\cos 30^\circ$$

$$= -\frac{1}{2}\sqrt{3}$$

c. $\tan 315^\circ = -\tan(360^\circ - 45^\circ)$

$$= -\tan 45^\circ$$

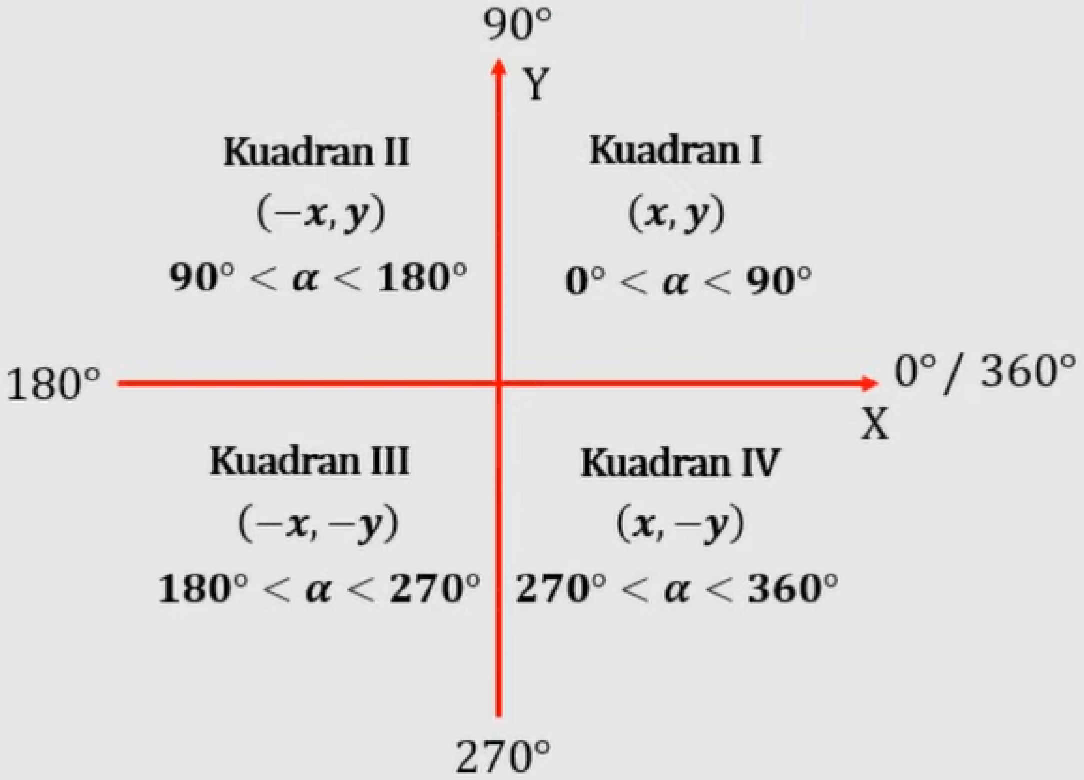
$$= -1$$

d. $\sec 300^\circ = \dfrac{1}{\cos 300^\circ}$

$$= \frac{1}{\cos(360^\circ - 60^\circ)}$$

$$= \frac{1}{\cos 60^\circ} = \frac{1}{\frac{1}{2}} = 1 \times \frac{2}{1} = 2$$

Letak sudut pada bidang cartesius



Tanda nilai trigonometri di berbagai kuadran

"Semua Sindikat Tangannya Kosong

I II III IV

Contoh 2

Tentukan nilai $\sin 120^\circ + \cos 210^\circ - \tan 225^\circ$

Contoh 2

Tentukan nilai $\sin 120° + \cos 210° - \tan 225°$

Penyelesaian

$$\begin{aligned}\sin 120° &= \sin(180° - 60°)\\ &= \sin 60°\\ &= \frac{1}{2}\sqrt{3}\end{aligned}$$

$$\begin{aligned}\tan 225° &= \tan(180° + 45°)\\ &= \tan 45°\\ &= 1\end{aligned}$$

$$\begin{aligned}\cos 210° &= -\cos(180° + 30°)\\ &= -\cos 30°\\ &= -\frac{1}{2}\sqrt{3}\end{aligned}$$

Contoh 2

Tentukan nilai sin 120° + cos 210° − tan 225°

Penyelesaian

$$\sin 120^\circ = \sin(180^\circ - 60^\circ) = \sin 60^\circ = \frac{1}{2}\sqrt{3}$$

$$\cos 210^\circ = -\cos(180^\circ + 30^\circ) = -\cos 30^\circ = -\frac{1}{2}\sqrt{3}$$

$$\tan 225^\circ = \tan(180^\circ + 45^\circ) = \tan 45^\circ = 1$$

$$\sin 120^\circ + \cos 210^\circ - \tan 225^\circ = \frac{1}{2}\sqrt{3} + \left(-\frac{1}{2}\sqrt{3}\right) - 1 = \frac{1}{2}\sqrt{3} - \frac{1}{2}\sqrt{3} - 1 = 0 - 1 = -1$$